Raghav Kochhar

raghavkochhar007@gmail.com - linkedin.com/in/raghavkochhar007 - github.com/Raghav-Kochhar
+1 778-955-3597 - Delta, BC, Canada - Canadian Permanent Resident

Summary

AI/ML Engineer with B.Tech in AI/ML building production ML on streaming data (Pub/Sub, Dataflow/Beam, Cloud Run, AlloyDB) for waste-to-energy truck monitoring (3K events/day), plus vision and LLM automation.

Experience

Artificial Intelligence and Machine Learning Engineer, Jaipur Robotics — August 2025 - Present

- Streaming and Data: Architected an Apache Beam (Python) streaming pipeline merging detection streams with plate normalization and confidence-ranked dedup (3K events/day, 4,800+ truck fleet, 13-min average end-to-end processing) into AlloyDB truck-session state.
- Computer Vision: Engineered waste-area polygon segmentation and alert-image processing services (1.6K boxes/day); supported LiDAR volume measurement (8.6K scans and 55M cubic meters/month).
- GenAI and Automation: Developed Gemini-based alert-visibility tracking (first/last-seen timestamps) and LLM-based alert classification across 9 plant sites (3.2K annotation tasks/month, under a minute to complete on average).
- Automated Analytics and Reporting: Automated truck-video and material-mixing analysis (1.3K analyses/day) feeding generated reports, reducing manual review effort.
- Platform: Shipped an async timelapse-video platform (FastAPI, GCS, Cloud Tasks, Streamlit) attaching video to 99% of sessions; modeled truck-tracking data in Django/AlloyDB.

Education

Vivekananda Institute of Professional Studies - Technical Campus — 2021 - 2025

Bachelor of Technology in Artificial Intelligence and Machine Learning (CGPA: 9.03/10).

Research

Efficient Adaptation of Lightweight LLMs (ICAAI 2025, Springer) - First author; accepted; B.Tech Major Project

- Benchmarked Baseline, T-Free, STELLA, and Combined on SST-2, MultiNLI, and SNLI (single RTX 3060 GPU).
- T-Free cut embedding parameters 74% with an accuracy trade-off; STELLA-tuned variant led at 0.784 SST-2 accuracy.

Health Helix: Connecting You to Better Health (COM-IT-CON 2024)

- Presented at the International Conference on Progressive Computational Intelligence (Taylor & Francis Group); architected a unified healthcare platform.
- Designed engagement workflows with automated AI alerts and scheduling integration aimed at reducing clinic no-show rates.

Projects

American Sign Language Interpreter

- Engineered real-time sign-language recognition with EfficientNetV2-S features and an LSTM head (MS-ASL).
- Shipped live webcam and video-upload inference via a Streamlit and OpenCV interface.

Speech Emotion Classifier

- Developed a seven-class LSTM emotion classifier on 40-dim MFCC features across 4 corpora (12K samples).
- Shipped a Streamlit UI for real-time and batch audio analysis with confidence-scored predictions.

Traffic & Pedestrian Analyzer

- Created a YOLOv8x traffic-monitoring system with pedestrian detection and KMeans car-color recognition.
- Deployed vehicle counting and analytics as an interactive Streamlit app.

Automaton: No-Code ML Workflow Platform

- Implemented a no-code ML platform for preprocessing, training, tuning, and benchmark visualization.

Technical Skills

- Languages: Python (pytest, mypy, ruff, uv), SQL, Bash.
- ML and Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, ONNX, Pandas, NumPy.
- LLMs and GenAI: LangChain, RAG pipelines, Prompt Engineering, Vertex AI.
- Computer Vision and Video: OpenCV, YOLOv8, FFmpeg.
- Web and APIs: FastAPI, Streamlit, Django, SQLAlchemy, Pydantic.
- Data Engineering: Apache Beam (Dataflow), Pub/Sub, BigQuery, PySpark, AlloyDB.
- Cloud and DevOps: GCP (Cloud Run, GKE, GCS, Cloud Tasks, Cloud Build), Docker, GitHub Actions.